

HUADIAN, China

WMO: 542730

Lat: 42.9783N

Lon: 126.7598E

Elev: 866

StdP: 14.24

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-18.2	-29.1	1.1	-23.0	-24.0	1.5	-17.6	20.8	10.6	17.6	10.0	0.7	70	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	87.4	71.7	85.1	71.1	82.9	69.9	76.2	83.3	74.6	81.3	73.1	79.4	6.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	131.4	80.8	72.5	124.4	78.9	71.0	117.9	77.3	40.5	83.5	38.9	81.5	37.4	79.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.2	13.8	-28.5	91.3	5.2	2.4	-32.2	93.1	-35.3	94.5	-38.2	95.8	-42.0	97.6
			-28.7	78.0	5.2	2.0	-32.4	79.5	-35.4	80.7	-38.4	81.8	-42.1	83.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	1.6	11.5	28.6	45.4	58.8	68.8	73.8	71.4	60.4	45.0	26.3	7.8
	DBStd	26.10	8.90	10.64	10.03	8.22	6.56	5.16	4.07	4.93	6.15	7.93	10.84	10.06
	HDD50	5653	1499	1079	663	185	8	0	0	0	4	196	712	1307
	HDD65	9138	1964	1499	1127	588	211	23	1	9	162	621	1161	1772
	CDD50	2647	0	0	0	47	281	565	737	662	315	40	0	0
	CDD65	660	0	0	0	1	19	137	274	206	23	0	0	0
	CDH74	4893	0	0	0	37	328	1097	1948	1297	179	7	0	0
	CDH80	1240	0	0	0	4	74	300	541	306	15	0	0	0
Wind	WSAvg	4.8	4.0	4.7	6.3	6.6	6.3	4.9	4.3	3.8	3.6	4.6	4.8	4.3
Precipitation	PrecAvg	29.3	0.4	0.4	0.6	1.7	2.6	4.3	7.4	6.3	2.8	1.5	0.9	0.4
	PrecMax	41.3	1.0	1.1	1.5	3.3	4.5	8.1	13.3	11.5	5.8	3.4	2.3	1.1
	PrecMin	20.9	0.0	0.0	0.0	0.5	0.7	1.5	3.3	1.7	0.7	0.2	0.2	0.0
	PrecStd	4.9	0.2	0.3	0.4	0.7	1.1	1.6	2.4	3.0	1.5	0.9	0.6	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	43.1	60.8	79.7	86.6	89.9	91.5	88.8	82.1	73.7	58.4	37.5
		MCWB	28.9	34.6	46.9	56.0	62.8	66.8	76.5	75.5	67.9	57.5	47.2	33.9
	2%	DB	26.2	38.0	53.6	72.5	81.8	86.3	87.7	85.8	78.7	68.3	51.3	33.1
		MCWB	22.9	32.1	41.2	53.0	59.8	66.2	74.5	74.0	65.7	54.6	43.4	29.7
	5%	DB	21.5	34.1	48.4	67.6	77.6	83.4	85.3	83.5	75.6	63.8	46.4	28.6
		MCWB	18.7	29.5	38.3	50.4	58.6	66.2	73.3	72.7	63.4	51.9	40.6	25.6
	10%	DB	17.1	30.1	44.0	62.6	73.5	80.6	83.0	81.0	72.8	59.4	41.6	24.0
		MCWB	14.7	26.2	35.9	47.9	57.0	66.0	72.5	71.5	61.5	50.0	36.7	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.0	35.8	47.9	59.6	66.6	72.5	79.0	77.9	71.3	59.4	48.4	34.7
		MCDB	31.9	41.3	59.8	75.0	81.0	81.6	87.2	85.9	78.8	70.4	56.6	36.5
	2%	WB	23.1	32.8	42.6	54.5	62.8	70.5	77.0	75.8	68.5	56.6	44.6	29.6
		MCDB	25.8	37.2	52.2	69.2	75.1	79.8	84.2	82.8	74.6	65.9	49.9	32.7
	5%	WB	18.8	29.6	39.0	51.5	60.4	68.9	75.4	74.4	66.0	53.9	41.1	25.8
		MCDB	21.5	33.9	47.6	65.5	72.7	78.1	82.3	80.9	72.5	61.2	46.0	28.6
	10%	WB	14.8	26.1	36.1	48.8	58.5	67.5	73.9	72.9	63.7	51.0	36.9	21.4
		MCDB	17.2	30.2	43.2	60.8	70.5	76.5	80.5	78.7	70.2	58.3	41.3	23.8
Mean Daily Temperature Range	5% DB	MDBR	25.0	24.7	19.7	22.4	21.7	18.3	15.2	15.9	22.2	22.9	18.7	21.9
		MCDBR	27.9	24.8	25.9	32.9	29.7	24.0	19.1	18.7	25.2	30.5	24.9	24.8
		MCWBR	24.7	19.3	15.1	15.6	11.0	8.4	7.6	8.2	12.4	16.4	16.9	21.8
	5% WB	MCDBR	27.3	23.7	25.0	29.2	24.0	18.8	16.1	15.5	20.1	26.4	23.3	24.4
		MCWBR	24.3	18.9	15.2	14.7	10.0	8.0	7.4	7.5	10.7	15.3	16.5	21.9
Clear-Sky Solar Irradiance	taub	0.364	0.416	0.499	0.590	0.642	0.672	0.632	0.551	0.485	0.521	0.442	0.381	
	taud	1.936	1.831	1.741	1.640	1.590	1.607	1.727	1.902	1.986	1.798	1.885	1.926	
	Ebn at Noon	225	233	229	220	213	208	214	228	233	197	193	203	
	Edh at Noon	39	52	64	77	83	82	72	58	50	53	41	36	
All-Sky Solar Radiation	RadAvg	751	1065	1360	1494	1695	1687	1518	1482	1365	980	656	582	
	RadStd	54	58	76	113	149	170	140	130	108	75	75	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page